HIDDEN COORDINATES AND SELECTION RULES IN THE FOURIER ENTROPY-INFLUENCE CONJECTURE

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ABSTRACT. For a Boolean function f and a coordinate i, the erasure entropy $E_i(f)$ is the conditional entropy of the event $i \in U$ given $U \setminus \{i\}$, where U is drawn from the spectral distribution $\hat{f}^2$. The Fourier entropy-influence conjecture is equivalent to every non-constant f having a coordinate with $E_i(f) \le c\,\mathrm{Inf}_i(f)$ for an absolute c. We study a family of monotone functions with a hidden coordinate: a variable that enters only through one point below a majority threshold. Its ratio E_i/Inf_i tends to infinity, yet the sum of the erasure entropies over all coordinates is at most twice the total influence, and at most five times the total influence after every other variable is replaced by a self-dual monotone function on fresh variables. These functions show that no rule that picks the coordinate of smallest influence, of largest influence, of largest or of smallest normalised level-two Fourier weight, or the variable lying in the most shortest minterms, can serve in the equivalence, even for monotone functions. We also show that the two singleton identities of monotone functions do not by themselves bound the erasure entropy, and we give the exact maxima of $\min_i E_i/\mathrm{Inf}_i$, H/I and $\sum_i E_i/I$ over all monotone functions of at most seven variables.

1. INTRODUCTION

For $f\colon \{0,1\}^n \to \{-1,1\}$ write $f = \sum_S \hat{f}(S)\chi_S$ with $\chi_S(x) = (-1)^{\sum_{j\in S} x_j}$, and let U be a random subset with $\Pr[U = S] = \mu(S) = \hat{f}(S)^2$. The Fourier entropy is $H(f) = H(U)$ and the total influence is $I(f) = \mathbb{E}|U| = \sum_i \mathrm{Inf}_i(f)$, where $\mathrm{Inf}_i(f) = \Pr[i \in U]$. The Fourier entropy-influence (FEI) conjecture of Friedgut and Kalai [2] asks for an absolute C with $H(f) \le C\,I(f)$ for every Boolean f. It is known for symmetric functions [7], for read-once formulas [6, 1] and for read-k decision trees with k fixed [12]; Han [4] bounds $H(f)$ by a multiple of $I(f)$ plus $\sum_i \mathrm{Inf}_i(f)\log_2(1/\mathrm{Inf}_i(f))$, and Hod [5] shows, with monotone functions, that the best constant exceeds 6.45.

The erasure entropy of f at i is

$$E_i(f) = H\big(\mathbf{1}[i \in U] \,\big|\, U \setminus \{i\}\big).$$

It is the part of the entropy drop from f to its two restrictions at x_i that is not controlled by a constant times Inf_i [4, Remarks 3.1–3.2], [3, (32)], and the FEI conjecture holds if and only if

$$G = \sup_f \min_{i\colon \mathrm{Inf}_i(f)>0} \frac{E_i(f)}{\mathrm{Inf}_i(f)}$$

is finite [8]. One direction is immediate from $\sum_i E_i \le H$. The other restricts f one coordinate at a time, each time at a coordinate with small ratio, and does not say which coordinate that is. This paper is about that choice.

Section 3 studies the functions

$$f_k(y,x) = \mathrm{MAJ}_k(y) \vee \big(x \wedge [y \ge y^0]\big), \qquad k \text{ odd},$$

where $y^0 \in \{0,1\}^k$ has $(k-1)/2$ ones and $[y \ge y^0]$ is the AND of the coordinates of y at which y^0 is 1. The variable x changes the value only at the single point $y = y^0$, so $\mathrm{Inf}_x(f_k) = 2^{-k}$,

and the ratio E_x/Inf_x tends to infinity (Proposition 5). Nevertheless the erasure entropies of all coordinates together are small:

Theorem 1. *For every odd k, $\sum_i E_i(f_k) \le 2\, I(f_k)$. If every y_j is replaced by a non-constant self-dual monotone function φ_j of its own block of fresh variables, the resulting monotone function F satisfies $E_x(F) = E_x(f_k)$, $E_\ell(F) < 4\,\mathrm{Inf}_\ell(F)$ at every relevant variable $\ell \ne x$, and $\sum_i E_i(F) \le 5\, I(F)$.*

The large ratio at x is paid for by the total influence: $E_x \le h_2(2^{-k})$, which is far below $I(f_k) \ge 1$. Two facts drive the proof. The parity of $|U|$ determines whether $i \in U$ once $U \setminus \{i\}$ is known, so every E_i is at most the binary entropy of the Fourier weight on even levels, and for f_k that weight is exactly 2^{-k} (Lemma 3). After the substitution, E_ℓ has an exact expression in terms of the spectrum of f_k (Proposition 6). Theorem 1 does not by itself give $H \le C\, I$ for these functions, since the passage from erasure entropies to H uses a good coordinate in every restriction and the family is not closed under restrictions; its point is that an unbounded ratio at one coordinate is compatible with a bounded sum.

Section 5 uses these functions against simple selection rules.

Theorem 2. *For each of the following rules there are monotone functions on which the coordinate it selects has E_i/Inf_i as large as we like, with ties broken in any way: the coordinate of smallest influence; of largest influence; of largest normalised level-two weight $W_i^{(2)}/\mathrm{Inf}_i$; of smallest normalised level-two weight; the variable lying in the most minterms of smallest size; the variable of smallest E_i/Inf_i among those lying both in a minterm of smallest size and in a maxterm of smallest size.*

Here $W_i^{(2)}(f) = \sum_{j \ne i} \hat{f}(\{i,j\})^2$. That the largest influence fails was shown in [9] by a different construction. For the rule of smallest normalised level-two weight, f_k itself is not a counterexample; there the hidden coordinate is made pivotal at two complementary points of the middle layer, which removes its level-two weight altogether (Proposition 9).

Section 6 shows that the identities $\hat{f}(\{i\})^2 = \mathrm{Inf}_i(f)^2$ and $\sum_i \mathrm{Inf}_i(f)^2 \le 1$, which hold for monotone f, are not enough to bound the erasure entropy: an exchangeable probability law on subsets of $[n]$ satisfies both and has $E_i/\Pr[i \in U] \ge \log_2(n)/288$. Section 7 gives the exact maxima of the three ratios $\min_i E_i/\mathrm{Inf}_i$, H/I and $\sum_i E_i/I$ over all monotone functions on at most seven variables, and the worst cases of the simple rules on at most six.

The last rule of Theorem 2 needs more than the hidden coordinate of f_k, which lies in no shortest maxterm. The rule is bounded by 2.6974 when a shortest minterm or maxterm is a single variable (Proposition 10), and on at most six variables its worst case equals $G_{\mathrm{mon}}(n)$. It fails because a variant of f_k has its hidden coordinate in both a shortest minterm and a shortest maxterm (Proposition 11), and substituting threshold functions of slightly different minterm and maxterm sizes leaves that coordinate the only variable in both, with almost the same ratio (Proposition 12).

2. Preliminaries

All logarithms are to base 2, $\eta = \log_2 e$, and h_2 is the binary entropy. For $p, q \ge 0$ put

$$\Phi(p,q) = (p+q)\, h_2\Big(\frac{q}{p+q}\Big) = p \log \frac{p+q}{p} + q \log \frac{p+q}{q}, \qquad \Phi(p,0) = \Phi(0,q) = 0.$$

Then

$$E_i(f) = \sum_{T \not\ni i} \Phi\big(\mu(T), \mu(T \cup \{i\})\big), \qquad \mathrm{Inf}_i(f) = \sum_{T \not\ni i} \mu(T \cup \{i\}). \tag{1}$$

Φ is concave and positively homogeneous of degree one, hence superadditive: $\sum_t \Phi(p_t, q_t) \le \Phi(\sum_t p_t, \sum_t q_t)$. It is nondecreasing in each argument, and $\Phi(p, q) \ge q \log(p/q)$ when $p \ge q$, since both of its terms are nonnegative and $(p+q)/q \ge p/q$. We use $h_2(t) \le t \log(e/t)$, which follows from $-(1-t)\ln(1-t) \le t$.

For monotone f (with $f = 1 - 2F$ for a monotone F, or $2F - 1$; squared coefficients do not depend on the choice), $|\hat{f}(\{i\})| = \mathrm{Inf}_i(f)$. A Boolean function is self-dual if $f(\bar{x}) = -f(x)$, where $\bar{x}$ is the complement of x; then $\hat{f}(S) = 0$ for every S of even size, and in particular $\hat{f}(\emptyset) = 0$.

Lemma 3. *Let $\varepsilon = \Pr[\,|U| \text{ even}]$. Then $E_i(f) \le h_2(\varepsilon)$ for every i, and $E_i(f) \le h_2(\mathrm{Inf}_i(f))$.*

Proof. Given $U \setminus \{i\}$, the parity of $|U|$ and the indicator of $i \in U$ determine each other, so $E_i = H(|U| \bmod 2 \mid U \setminus \{i\}) \le H(|U| \bmod 2) = h_2(\varepsilon)$. The second bound drops the conditioning. □

3. The hidden coordinate

Fix an odd k, let $y^0 \in \{0,1\}^k$ have $(k-1)/2$ ones, and let f_k be as in the introduction, written in ± 1 values with $+1$ for true. Put

$$\delta = 2^{-k}, \qquad a = \binom{k-1}{(k-1)/2} 2^{1-k},$$

so that a is the influence of every variable of MAJ_k. For $k = 1$ the vector y^0 is empty and $f_1 = y \vee x$.

Lemma 4. (1) *$f_k = \mathrm{MAJ}_k + (1+x)\mathbf{1}[y = y^0]$, where x is the ± 1 value of the hidden variable.*
(2) *$\mathrm{Inf}_x(f_k) = \delta$; $\mathrm{Inf}_{y_j}(f_k) = a + \delta$ if $y^0_j = 1$ and $a - \delta$ if $y^0_j = 0$; $I(f_k) = ka \ge 1$.*
(3) *The Fourier weight of f_k on sets of even size is exactly δ.*

Proof. If $y \ge y^0$ and $y \ne y^0$ then $|y| \ge (k+1)/2$, so the clause $x \wedge [y \ge y^0]$ changes the value only at $y = y^0$, where $\mathrm{MAJ}_k = -1$ and $f_k = x$; this is (1). The derivative in x is $\mathbf{1}[y = y^0]$, so $\mathrm{Inf}_x = \Pr[y = y^0] = \delta$. Setting $f_k(1, y^0) = 1$ creates a pivotal edge in direction j when $y^0_j = 1$, since $\mathrm{MAJ}_k(y^0 - e_j) = -1$, and removes one when $y^0_j = 0$, since $\mathrm{MAJ}_k(y^0 + e_j) = 1$; each edge carries influence δ. There are $(k-1)/2$ coordinates of the first kind and $(k+1)/2$ of the second, so $I(f_k) = \delta + ka - \delta = ka$. As a is the largest atom of the binomial law on $\{0, \dots, k-1\}$, $a \ge 1/k$. For (3): MAJ_k is self-dual, so its spectrum lies on odd sets not containing x. The coefficients of $(1+x)\mathbf{1}[y = y^0]$ all have modulus δ, one at each T and one at each $T \cup \{x\}$, $T \subseteq [k]$; the 2^k sets of even size among them carry no weight of MAJ_k and have total weight $2^k \delta^2 = \delta$. □

Proof of Theorem 1 for f_k. By Lemmas 3 and 4, $\sum_i E_i(f_k) \le (k+1) h_2(\delta)$, and $I(f_k) \ge 1$. For $k = 1$, $f_1 = y \vee x$ has μ uniform on four sets, so $E_x = E_y = 1$ and $\sum_i E_i = 2 = 2I$. For $k = 3$, $I = 3/2$ and $4h_2(1/8) < 2.18 < 3$. For odd $k \ge 5$, $(k+1)h_2(\delta) \le (k+1)(k+\eta)2^{-k}$, which is < 1.21 at $k = 5$ and decreases, since its ratio from k to $k+2$ is $\frac{1}{4}\frac{k+3}{k+1}\frac{k+2+\eta}{k+\eta} < 1$. □

Proposition 5. *For odd k large enough,*

$$\frac{E_x(f_k)}{\mathrm{Inf}_x(f_k)} \;\ge\; \Big(\frac{1}{2} - \frac{4k}{(k-3)^2}\Big)\Big(\frac{k-1}{4} - 2\log k\Big).$$

Proof. Write $g = \mathrm{MAJ}_k + \mathbf{1}[y = y^0]$ for the average of f_k over x. By (1), $E_x = \sum_T \Phi(\hat{g}(T)^2, \delta^2)$ over $T \subseteq [k]$, and $\mathrm{Inf}_x = 2^k \delta^2$, so E_x/Inf_x is the average of $\delta^{-2}\Phi(\hat{g}(T)^2, \delta^2)$ over uniform T. For $|T| = r$ odd, $\widehat{\mathrm{MAJ}_k}(T)$ has modulus

$$b_r = a \binom{(k-1)/2}{(r-1)/2} \Big/ \binom{k-1}{r-1}.$$

Indeed, for $j \in T$ it equals the coefficient at $T \setminus \{j\}$ of the derivative in y_j, the indicator of the middle layer of the other $2m = k-1$ coordinates, and $\sum_S \mathbb{E}[\chi_S \mid |z| = m]\, t^{|S|} = (1-t)^m(1+t)^m = (1-t^2)^m$ gives $\mathbb{E}[\chi_S \mid |z| = m] = (-1)^s\binom{m}{s}/\binom{2m}{2s}$ for $|S| = 2s$. If $(k-1)/4 \le r-1 \le 3(k-1)/4$ then $\binom{m}{(r-1)/2} \ge 2^{(k-1)/8}$, and with $a \ge 1/k$ and $\binom{k-1}{r-1} \le 2^{k-1}$ we get $b_r/\delta \ge 2^{(k-1)/8+1}/k \ge 2$ for large k. Then $|\hat{g}(T)| \ge b_r - \delta \ge b_r/2$ and $\delta^{-2}\Phi(\hat{g}(T)^2, \delta^2) \ge \log(\hat{g}(T)^2/\delta^2) \ge (k-1)/4 - 2\log k$. The set of such T has probability at least $\frac{1}{2} - \Pr[\,||T| - k/2| > (k-3)/4\,] \ge \frac{1}{2} - \frac{4k}{(k-3)^2}$ by Chebyshev's inequality. □

The ratio is 3.52, 5.78 and 9.27 at $k = 3, 9, 17$.

4. Substitution

Let $\varphi_1, \dots, \varphi_k$ be non-constant self-dual monotone functions on disjoint blocks of fresh variables, and $F(z, x) = f_k(\varphi_1(z^1), \dots, \varphi_k(z^k), x)$. Write μ for the spectral distribution of f_k and δ_j for $\mathrm{Inf}_{y_j}(f_k)$. For a variable ℓ of block j let $\tau = \mathrm{Inf}_\ell(\varphi_j) = |\hat{\varphi}_j(\{\ell\})|$.

Proposition 6. *With $p_T = \mu(T)$ and $q_T = \mu(T \cup \{y_j\})$ for T a subset of the coordinates of f_k other than y_j,*

$$\mathrm{Inf}_\ell(F) = \tau\delta_j, \qquad E_\ell(F) = \sum_T \Phi(p_T, \tau^2 q_T).$$

Moreover $E_x(F) = E_x(f_k)$, $\mathrm{Inf}_x(F) = \delta$, and the weight of F on sets of even size is δ.

Proof. Since $\hat{\varphi}_j(\emptyset) = 0$, expanding $f_k = \sum_T \hat{f}_k(T)\prod_{j\in T} y_j$ and substituting gives $\hat{F}(S) = \hat{f}_k(T)\prod_{j\in T}\hat{\varphi}_j(S_j)$, where $S_j = S \cap \mathrm{block}_j$ and T consists of x if $x \in S$ and the j with $S_j \neq \emptyset$. Fix the parts of a set $V \not\ni \ell$ outside block j. If $V_j \neq \emptyset$, the sets V and $V \cup \{\ell\}$ have the same T, and their block-j parts differ in size by one; as φ_j is self-dual, one of the two coefficients vanishes and the pair contributes $\Phi(\cdot, 0) = 0$. If $V_j = \emptyset$, the pair is $(T, T \cup \{y_j\})$ with factors 1 and τ^2, times a common factor $\prod_{i\in T}\hat{\varphi}_i(S_i)^2$. By homogeneity of Φ and $\sum_{S_i\neq\emptyset}\hat{\varphi}_i(S_i)^2 = 1$ (Parseval and $\hat{\varphi}_i(\emptyset) = 0$), summing over the other blocks leaves $\Phi(p_T, \tau^2 q_T)$. The influence follows in the same way from $\sum_{S\ni\ell}\hat{\varphi}_j(S)^2 = \tau$. For x, the pairs $(V, V \cup \{x\})$ correspond to $(T, T \cup \{x\})$ with a common factor, so E_x and Inf_x are unchanged. Each nonempty S_i has odd size, so $|S| \equiv |T| \pmod 2$ and the even weight is that of f_k. □

Corollary 7. $E_\ell(F) \le \min\big(h_2(\delta), h_2(\tau^2\delta_j)\big)$.

Proof. The first bound is Lemma 3 with Proposition 6. For the second, superadditivity gives $E_\ell(F) \le \Phi(\sum_T p_T, \tau^2\sum_T q_T) = \Phi(1-\delta_j, \tau^2\delta_j) \le \Phi(1-\tau^2\delta_j, \tau^2\delta_j)$, using that Φ is nondecreasing in its first argument. □

Proof of Theorem 1 for F. Let $k \ge 3$. Then $\delta \le 1/(2k)$, so $\delta_j \ge a - \delta \ge 1/(2k)$. Put $r = \sqrt{\delta/\delta_j}$. If $\tau \ge r$, Corollary 7 and $h_2(t) \le t\log(e/t)$ give

$$\frac{E_\ell}{\mathrm{Inf}_\ell} \le \frac{h_2(\delta)}{r\delta_j} \le r\log\frac{e}{\delta}.$$

If $\tau \le r$, they give $E_\ell/\mathrm{Inf}_\ell \le \tau\log(e/(\tau^2\delta_j))$, whose derivative in τ is $\log(1/(\tau^2\delta_j)) - \eta > 0$, because $\tau^2\delta_j \le \delta \le 1/8 < 1/e$; so it is at most its value $r\log(e/\delta)$ at $\tau = r$. Hence

$$\frac{E_\ell(F)}{\mathrm{Inf}_\ell(F)} \le \sqrt{\delta/\delta_j}\,\log(e/\delta) \le B_k := \sqrt{2k2^{-k}}\,(k+\eta).$$

Here $B_3 < \frac{9}{4}\sqrt{3} < 4$, and $B_{k+2}/B_k = \frac{1}{2}\sqrt{1+2/k}\,(1+2/(k+\eta)) \le \frac{1}{2}\sqrt{5/3}\cdot\frac{3}{2} < 1$. For $k = 1$, $\delta = \delta_1 = 1/2$ and $E_\ell/\mathrm{Inf}_\ell \le h_2(\tau^2/2)/(\tau/2) \le \tau\log(2e/\tau^2) \le 2\sqrt{2/e}\,\eta < 3$.

For the sum, $I(F) = \delta + \sum_j \delta_j I(\varphi_j)$, and $I(\varphi_j) \geq \mathrm{Var}(\varphi_j) = 1$, so $I(F) \geq \delta + \sum_j \delta_j = I(f_k) \geq 1$. Therefore

$$\sum_i E_i(F) = E_x(F) + \sum_\ell E_\ell(F) \leq h_2(\delta) + 4\sum_j \delta_j I(\varphi_j) \leq I(F) + 4I(F). \qquad \square$$

Remark 8. The bound $E_\ell < 4\mathrm{Inf}_\ell$ holds at every variable of every block, whatever the sizes of the blocks; in particular for majority blocks of any size and for iterated majorities. Since x keeps the same E_x and Inf_x, the ratio at x still tends to infinity with k.

5. Selection rules

Proof of Theorem 2. Smallest influence. In f_k, $\mathrm{Inf}_x = \delta < a - \delta \leq \mathrm{Inf}_{y_j}$ for $k \geq 3$, so the rule picks x, and Proposition 5 applies.

Largest normalised level-two weight. MAJ_k has no weight on sets of size two, so every coefficient of f_k at a set of size two is a coefficient of $(1+x)\mathbf{1}[y = y^0]$, of modulus δ. Thus $W_i^{(2)} = k\delta^2$ for every coordinate, and maximising $W_i^{(2)}/\mathrm{Inf}_i$ is minimising Inf_i.

Largest influence. Substitute $\varphi_j = \mathrm{MAJ}_{2s+1}$ for every j. Then $\tau = \binom{2s}{s}4^{-s} \leq (s+1)^{-1/2}$ (the ratio from s to $s+1$ is $\frac{2s+1}{2s+2}$, and $(\frac{2s+1}{2s+2})^2 \leq \frac{s+1}{s+2}$). For $s+1 > 4^k$ every variable of a block has influence $\tau\delta_j < \delta$, so x is the unique coordinate of largest influence, and by Proposition 6 its ratio is that of f_k.

Most shortest minterms. Substitute $\varphi_j = \mathrm{MAJ}_3$. The minterms of F are x together with a minterm of MAJ_3 in each block j with $y_j^0 = 1$, of size $1 + 2 \cdot \frac{k-1}{2} = k$, and minterms of MAJ_3 in $(k+1)/2$ of the blocks, of size $k+1$. So every shortest minterm contains x, while a variable of a block lies in two of the three minterms of MAJ_3 and hence in fewer of them. The rule picks x.

Smallest normalised level-two weight. This is Proposition 9 below.

Smallest ratio among the variables in a shortest minterm and a shortest maxterm. This is Proposition 12 below. $\square$

For f_k the hidden coordinate has the same level-two weight as every other coordinate, which is not the smallest normalised value. The next construction removes it.

Let $k \geq 6$ be even, let $y^0 \in \{0,1\}^k$ have $k/2$ ones and $y^1 = \bar{y}^0$. Let $B(y) = 1$ if $|y| \geq k/2$ and 0 otherwise, and define $f'_k(y, x) = B(y)$ except that $f'_k(y, x) = x$ when $y \in \{y^0, y^1\}$. Lowering a point of the middle layer from 1 to 0 preserves monotonicity, so f'_k is monotone.

Proposition 9. *$W_x^{(2)}(f'_k) = 0$ and $W_{y_j}^{(2)}(f'_k) > 0$ for every j, so x is the unique coordinate of smallest normalised level-two weight;* $\mathrm{Inf}_x = 2^{1-k}$, $\mathrm{Inf}_{y_j} = \binom{k}{k/2}2^{-k}$ *for every j; and for even k large enough*

$$\frac{E_x(f'_k)}{\mathrm{Inf}_x(f'_k)} \;\geq\; \Big(1 - \frac{8}{k}\Big)\Big(\frac{k}{4} - 4 - 2\log(k+1)\Big).$$

Proof. With $P = \mathbf{1}[y \in \{y^0, y^1\}]$ we have $f'_k = B - (1-x)P$ in ± 1 values, the derivative in x is P, and the average over x is $g = B - P$. Put $\delta = 2^{1-k}$. Since $\chi_T(y^1) = (-1)^{|T|}\chi_T(y^0)$, $\hat{P}(T)$ vanishes for odd $|T|$ and has modulus δ for even $|T|$. So $\mathrm{Inf}_x = \Pr[y \in \{y^0, y^1\}] = \delta$ and the coefficient of f'_k at $\{x, y_j\}$, which is $\hat{P}(\{y_j\})$, is zero: $W_x^{(2)} = 0$. At y^0 and at y^1 the change creates or removes a pivotal edge in direction j according to the jth bit, and the two bits are opposite, so $\mathrm{Inf}_{y_j}(f'_k) = \mathrm{Inf}_j(B) = \binom{k}{k/2}2^{-k} =: a_0$.

In ± 1 values, B is the sum of an odd function, the sign of $|y| - k/2$ with 0 on the middle layer, and the indicator of the middle layer. Only the latter has even coefficients, and by the

generating function of Proposition 5, for even $r = |T|$,

$$|\hat{B}(T)| = a_0 \binom{k/2}{r/2} \Big/ \binom{k}{r}. \tag{2}$$

At $r = 2$ this is $a_0/(k-1)$, and $a_0/((k-1)\delta) = \binom{k}{k/2}/(2(k-1))$ equals 2 at $k = 6$ and increases. So $|\hat{f}'_k(\{y_j, y_l\})| \geq |\hat{B}(\{y_j, y_l\})| - \delta > 0$ and $W^{(2)}_{y_j} > 0$.

The derivative spectrum is uniform over the even subsets T of $[k]$, and E_x/Inf_x is the average of $\delta^{-2}\Phi(\hat{g}(T)^2, \delta^2)$. For even $r \in [k/4, 3k/4]$, (2) with $\binom{k/2}{r/2} \geq 2^{k/8}$, $a_0 \geq 1/(k+1)$ and $\binom{k}{r} \leq 2^k$ gives $|\hat{B}(T)| \geq 2^{-7k/8}/(k+1) \geq 2\delta$ for large k, hence $|\hat{g}(T)| \geq |\hat{B}(T)|/2$ and $\delta^{-2}\Phi(\hat{g}(T)^2, \delta^2) \geq k/4 - 4 - 2\log(k+1)$. By Chebyshev's inequality, conditioned on $|T|$ even, $|T| \in [k/4, 3k/4]$ with probability at least $1 - 8/k$. □

At $k = 18$ the ratio at x is 14.28, while the smallest ratio over all coordinates is 1.60.

Shortest minterms and shortest maxterms. In each of these functions the hidden coordinate does not lie both in a shortest minterm and in a shortest maxterm: in f_k and its substitutions it lies in no shortest maxterm, and in f'_k in no shortest minterm. For monotone f let $A(f)$ be the set of variables lying in some shortest minterm and $B(f)$ the set lying in some shortest maxterm. Every minterm meets every maxterm, so $A(f) \cap B(f)$ is not empty. Consider the rule that chooses a variable of $A(f) \cap B(f)$ minimising E_i/Inf_i. It is bounded when a shortest minterm or a shortest maxterm is a single variable, but not in general.

Proposition 10. *Let $f = x \wedge g$, where g is a monotone function of the other variables that is not identically false, and let $p = \Pr[g \text{ true}]$. Then $\mathrm{Inf}_x(f) = p$ and*

$$\frac{E_x(f)}{\mathrm{Inf}_x(f)} = 2(1-p) + \frac{\Phi\big((1-p)^2, p^2\big)}{p} \;\leq\; 4(1-p).$$

The middle expression tends to 2 as $p \to 0$, equals 2 at $p = 1/2$, and its maximum over $p \in (0, 1]$ is 2.697377, attained near $p = 0.1994$.

Proof. Write x and g in ± 1 values with -1 for true. Then $f = \frac{1+g}{2} + x\,\frac{1-g}{2}$, so for T not containing x, $\hat{f}(T) = \frac{1}{2}(\mathbf{1}[T = \emptyset] + \hat{g}(T))$ and $\hat{f}(T \cup \{x\}) = \frac{1}{2}(\mathbf{1}[T = \emptyset] - \hat{g}(T))$. As $\hat{g}(\emptyset) = 1 - 2p$, we get $\mu(\emptyset) = (1-p)^2$, $\mu(\{x\}) = p^2$, and $\mu(T) = \mu(T \cup \{x\}) = \hat{g}(T)^2/4$ for $T \neq \emptyset$. Since $\Phi(c, c) = 2c$, (1) gives

$$E_x = \Phi\big((1-p)^2, p^2\big) + \tfrac{1}{2}\big(1 - (1-2p)^2\big), \qquad \mathrm{Inf}_x = p^2 + \tfrac{1}{4}\big(1 - (1-2p)^2\big) = p,$$

and the second term of E_x is $2p(1-p)$. For the bound, $\Phi(a, b) = (a+b)h_2(b/(a+b)) \leq 2\sqrt{ab}$ by the inequality $h_2(t) \leq 2\sqrt{t(1-t)}$, which follows from $h_2(t) \leq (4t(1-t))^{1/\ln 4}$ [11]; here $2\sqrt{ab} = 2p(1-p)$. □

If a shortest maxterm of a monotone f is a single variable x, then $f = x \wedge g$ with g the restriction $x = 1$, and every shortest maxterm is a single variable of this kind. So every variable of $B(f)$, and hence of $A(f) \cap B(f)$, has $E_i/\mathrm{Inf}_i < 2.6974$. The dual $f^\dagger(z) = \neg f(\bar{z})$ exchanges minterms and maxterms and has the same squared Fourier coefficients, so the same holds when a shortest minterm is a single variable. In these cases any variable of $A \cap B$ will do.

In general a variable of $A \cap B$ can be bad. For odd $k \geq 3$ let $m = (k-1)/2$, let y^0 have ones exactly at $y_1, \ldots, y_m$, put $y^1 = y^0 + e_{m+1}$, let M be MAJ_k with its value at y^1 changed to false, and

$$\tilde{f}_k(y, x) = M(y) \vee \big(x \wedge [y \geq y^0]\big).$$

The point y^1 lies in the lowest true layer of MAJ_k, so M and $\tilde{f}_k$ are monotone.

Proposition 11. *The variable x lies in a shortest minterm and in a shortest maxterm of $\tilde{f}_k$, $\mathrm{Inf}_x(\tilde{f}_k) = 2^{1-k}$, and for odd k large enough*

$$\frac{E_x(\tilde{f}_k)}{\mathrm{Inf}_x(\tilde{f}_k)} \;\geq\; \Big(\frac{1}{2} - \frac{4(k-1)}{(k-5)^2}\Big)\Big(\frac{k-1}{4} - 2\log k\Big).$$

Proof. A true point has weight at least $m+1$: with x false it needs $M(y)$, so $|y| \geq m+1$, and with x true it needs $y \geq y^0$ or $M(y)$. Hence $\{x, y_1, \dots, y_m\}$ is a shortest minterm. Every point of weight $m+2$ is true: with x false, $|y| = m+2$ and $M(y)$ holds; with x true, $|y| = m+1$ and either $y = y^1 \geq y^0$ or $M(y)$ holds. The point with x false and $y = y^1$ is false and has weight $m+1$, so the variables at which it is 0, among them x, form a shortest maxterm.

In ± 1 values with $+1$ for true, $\tilde{f}_k = \mathrm{MAJ}_k + (1+x)\mathbf{1}[y = y^0] - (1-x)\mathbf{1}[y = y^1]$. The derivative in x is $D = \mathbf{1}[y = y^0] + \mathbf{1}[y = y^1]$ and the average over x is $\bar{g} = \mathrm{MAJ}_k + \mathbf{1}[y = y^0] - \mathbf{1}[y = y^1]$. With $\delta = 2^{-k}$, $\hat{D}(T) = \delta(\chi_T(y^0) + \chi_T(y^1))$ vanishes if $y_{m+1} \in T$ and has modulus 2δ otherwise. So $\mathrm{Inf}_x = 2^{k-1}(2\delta)^2 = 2^{1-k}$, and E_x/Inf_x is the average of $(2\delta)^{-2}\Phi(\hat{\bar{g}}(T)^2, 4\delta^2)$ over the 2^{k-1} sets $T \subseteq [k] \setminus \{m+1\}$. For these T, $\chi_T(y^0) = \chi_T(y^1)$, the two indicators cancel, and $|\hat{\bar{g}}(T)| = b_r$ for $|T| = r$ odd, with b_r as in the proof of Proposition 5. If $(k-1)/4 \leq r-1 \leq 3(k-1)/4$, that proof gives $b_r/(2\delta) \geq 2^{(k-1)/8}/k \geq 1$ for large k, hence $(2\delta)^{-2}\Phi(b_r^2, 4\delta^2) \geq \log(b_r^2/(4\delta^2)) \geq (k-1)/4 - 2\log k$. Now $|T|$ is binomial with parameters $k-1$ and $\frac{1}{2}$; it is odd with probability $\frac{1}{2}$, and by Chebyshev's inequality $\big||T| - \frac{k-1}{2}\big| > \frac{k-5}{4}$ with probability at most $4(k-1)/(k-5)^2$. Otherwise $(k+3)/4 \leq |T| \leq (3k-7)/4$, which is the required range. □

At $k = 3, 5, \dots, 13$ the ratio at x is 1.80, 2.56, 3.17, 3.90, 4.70 and 5.55, while the smallest ratio over $A(\tilde{f}_k) \cap B(\tilde{f}_k)$ is the smallest over all coordinates, 1.00 at $k = 3$ and 0.0060 at $k = 13$. So $\tilde{f}_k$ does not defeat the rule. A substitution removes the other variables from $A \cap B$.

For $t \geq 1$ let T_t be the function of $2t$ variables that is true when at least t of them are true, and T_t' the one that is true when at least $t+1$ are. Their shortest minterms have sizes t and $t+1$, and their shortest maxterms sizes $t+1$ and t. Let $F_{k,t}$ be obtained from $\tilde{f}_k$ by replacing y_j with T_t of a block of fresh variables for $j \leq m$ and with T_t' of a block of fresh variables for $j > m$.

Proposition 12. *$A(F_{k,t}) \cap B(F_{k,t}) = \{x\}$, and for each k, $E_x(F_{k,t})/\mathrm{Inf}_x(F_{k,t}) \to E_x(\tilde{f}_k)/\mathrm{Inf}_x(\tilde{f}_k)$ as $t \to \infty$. Hence the rule that chooses the variable of smallest E_i/Inf_i in $A \cap B$ has unbounded ratio on monotone functions.*

Proof. If each y_j of a monotone g is replaced by a non-constant monotone φ_j of its own block, the minterms of the result are the sets $\bigcup_{j \in S} R_j$, together with x when $x \in S$, where S is a minterm of g and each R_j is a minterm of φ_j; the same holds for maxterms. So a shortest minterm of the result comes from a minterm S of g minimising $\sum_{j \in S} a_j$, where $a_x = 1$ and a_j is the size of a shortest minterm of φ_j, and likewise for maxterms.

Write $S_0 = \{y_1, \dots, y_m\}$ and $R = \{y_{m+2}, \dots, y_k\}$. The minterms of $\tilde{f}_k$ are $\{x\} \cup S_0$ and the sets of $m+1$ variables y_j other than $S_0 \cup \{y_{m+1}\}$: a true point either has $y \geq y^0$ and x true or satisfies M, and M is true exactly at the points of weight at least $m+1$ other than y^1. The first has $\sum a_j = 1 + mt$. Any other contains $i \leq m$ variables of S_0 and $m+1-i \geq 1$ others, and has $\sum a_j = (m+1)t + (m+1-i) > 1 + mt$. The maximal false points of $\tilde{f}_k$ are the point with x false and $y = y^1$, and the points with x true and $|y| = m$, $y \neq y^0$; the maxterms are therefore $\{x\} \cup R$ and the sets of $m+1$ variables y_j other than $R \cup \{y_{m+1}\}$. The first has $\sum b_j = 1 + mt$, and every other contains a variable of S_0 and has $\sum b_j \geq (m+1)t + 1$. Hence every shortest minterm of $F_{k,t}$ consists of x and variables of the blocks of S_0, every shortest maxterm consists of x and variables of the blocks of R, and $A \cap B = \{x\}$.

For the ratio let μ_j be the probability that φ_j is true, $\sigma_j^2 = \mu_j(1-\mu_j)$, and expand $\tilde{f}_k$, in ± 1 values, in the orthonormal basis $\prod_{j\in S}(y_j-\mu_j)/\sigma_j$ (times x when $x \in S$) of the product measure with $\Pr[y_j = 1] = \mu_j$ and x uniform; call the coefficients $c_\mu(S)$. Each $\psi_j = (\varphi_j - \mu_j)/\sigma_j$ has mean 0 and variance 1 under the uniform measure on its block, so $\sum_{R_j\neq\emptyset} \hat{\psi}_j(R_j)^2 = 1$, and the coefficient of $F_{k,t}$ at the union of $S \cap \{x\}$ and sets $R_j \neq \emptyset$, $j \in S \setminus \{x\}$, is $c_\mu(S)\prod_j \hat{\psi}_j(R_j)$. So the spectral sample of $F_{k,t}$ is obtained by drawing S with probability $c_\mu(S)^2$ and then each R_j independently. It contains x exactly when S does, it determines $S \setminus \{x\}$, and given $S \setminus \{x\}$ the R_j do not depend on whether $x \in S$. Hence $E_x(F_{k,t})$ and $\mathrm{Inf}_x(F_{k,t})$ are the erasure entropy and the influence of x for the law $c_\mu(S)^2$. Here $\mu_j = P_t$ for $j \leq m$ and $1 - P_t$ for $j > m$, with $P_t = \Pr[\mathrm{Bin}(2t, \frac{1}{2}) \geq t] \to \frac{1}{2}$. The coefficients $c_\mu(S)$ are continuous in $\mu \in (0,1)^k$ and Φ is continuous, so both quantities converge to their values at $\mu = \frac{1}{2}$, which are $E_x(\tilde{f}_k)$ and $\mathrm{Inf}_x(\tilde{f}_k) > 0$. The last statement follows from Proposition 11. □

Already $t = 1$, where T_1 and T_1' are the OR and the AND of two variables, gives a growing ratio at the forced coordinate, on $2k+1$ variables. By the proof above it is the ratio of $\tilde{f}_k$ under the product measure with bias $\frac{3}{4}$ on $y_1, \ldots, y_m$ and $\frac{1}{4}$ on $y_{m+1}, \ldots, y_k$. There D and the average over x have coefficients that depend only on $|S \cap \{y_1, \ldots, y_m\}|$ and $|S \cap \{y_{m+2}, \ldots, y_k\}|$, so the ratio is a double sum over these two numbers and can be computed exactly. It is 1.05, 1.88, 2.57, 3.12, 3.59, 4.12, 4.70 and 5.30 for $k = 3, 5, \ldots, 17$, and 13.83, 29.31, 45.33 and 52.34 at $k = 41, 81, 121, 141$, an increase of about 0.39 per unit of k; for $\tilde{f}_k$ the values at these four k are 18.62, 37.39, 57.09 and 66.98. The sequence is not monotone (it falls from 52.59 at $k = 133$ to 50.32 at $k = 135$). Under a bias the coefficients of the majority near its threshold are Krawtchouk-type sums that change sign, and the proof of Proposition 11 does not carry over. For $t = 1$ the ratio is $(1 + o(1))(k-1)\log_2 \frac{4}{3}$, which is $(1 + o(1))\log_2(1/\mathrm{Inf}_x)$ and so asymptotic to its largest possible value [10]; the proof writes these sums as integrals over the unit circle. For fixed $t \geq 2$ we do not know a proof that the ratio is unbounded. At $k = 7$, $t = 1$ the smallest ratio over all coordinates is 2.15, and at $k = 17$, $t = 20$ the forced ratio is 9.59. On monotone functions of at most six variables the rule does no worse than $G_{\mathrm{mon}}(n)$ (Section 7).

6. The singleton identities do not suffice

For monotone f, $\mu(\{i\}) = \mathrm{Inf}_i(f)^2 = \Pr[i \in U]^2$ for every i, and hence $\sum_i \Pr[i \in U]^2 \leq 1$. The following law satisfies both identities, is invariant under permutations, and has unbounded erasure ratio. It is not the spectral distribution of a Boolean function, so it shows only that any proof must use more of the structure.

Proposition 13. *For $n \geq 100$ let $p = 1/n$, let ρ be the law of a random subset of $[n]$ containing each element independently with probability p, conditioned on having at least two elements, let $d = \rho[i \in U]$, $b = 1/4$, and $\delta = (1 - \sqrt{1 - 4bd})/2$. The law μ that gives mass $b\rho(S)$ to each S with $|S| \geq 2$, mass δ^2 to each singleton and the rest to $\emptyset$ is a probability law with $\mu(\{i\}) = \Pr_\mu[i \in U]^2$, and*

$$\frac{H(\mathbf{1}[i \in U] \mid U \setminus \{i\})}{\Pr_\mu[i \in U]} \;\geq\; \frac{\log n}{288}.$$

Proof. Let $q = \Pr[\mathrm{Bin}(n,p) \geq 2] \geq \Pr[\mathrm{Bin}(n,p) = 2] = \frac{n-1}{2n}(1-\frac{1}{n})^{n-2} \geq \frac{1}{12}$, using $(1-1/n)^{n-2} \geq e^{-1} > 1/3$. Then $d \leq p/q \leq 12/n$, $bd \leq 3/n < 1/4$, so δ is real, $\delta \leq 2bd \leq 6/n$ and $n\delta^2 \leq 36/n \leq 0.36$; the mass of $\emptyset$ is at least $1 - \frac{1}{4} - 0.36 > 0$. By the choice of δ, $\Pr_\mu[i \in U] = \delta^2 + bd = \delta$. In (1) keep only the $T \not\ni i$ with $|T| \geq 2$: both T and $T \cup \{i\}$ lie in the product part, with masses in ratio $(1-p) : p$, so each pair contributes $(\mu(T) + \mu(T \cup \{i\}))h_2(p)$, and the pairs have total

mass $(b/q)\Pr[\mathrm{Bin}(n-1,p) \geq 2] \geq \frac{1}{4} \cdot \frac{1}{12}$, the last factor by the same estimate as for q. With $\delta \leq 6/n$ and $h_2(1/n) \geq (\log n)/n$ this gives $E_i/\delta \geq n\, h_2(1/n)/288 \geq (\log n)/288$. □

The exact ratio is 4.59, 5.95, 7.34 and 8.73 at $n = 10^2, 10^3, 10^4, 10^5$.

7. Monotone functions on few variables

Let $G_{\mathrm{mon}}(n)$ be the maximum of $\min_{i:\mathrm{Inf}_i>0} E_i/\mathrm{Inf}_i$ over non-constant monotone functions on n variables, and similarly for H/I and $\sum_i E_i/I$. Monotone functions on n variables are the pairs (f_0, f_1) of monotone functions on $n-1$ variables with $f_0 \leq f_1$; there are 7,828,354 of them for $n = 6$ and 2,414,682,040,998 for $n = 7$, the Dedekind numbers. For $n \leq 6$ the program `census_mono.py` evaluates every labelled function. The program `mono_census.c` lets the first function of each pair range over one representative of each orbit of the symmetric group on the first $n-1$ variables, weighted by the orbit size, and checks that the weighted count is the Dedekind number; at $n = 6$ its three maxima and its worst cases for the rules of largest and smallest influence agree with those of `census_mono.py` to 10^{-12}, although `mono_census.c` breaks ties in favour of a rule and `census_mono.py` against it. It gives the row $n = 7$. Table 1 gives the results.

n	monotone functions	$G_{\mathrm{mon}}(n)$	$\max H/I$	$\max \sum_i E_i/I$
2	6	2	2	2
3	20	2.672488983973	2.955889582252	2.672488983973
4	168	2.672488983973	3.402475551199	2.704153205629
5	7,581	2.695887520153	3.623740399702	2.883566415773
6	7,828,354	2.944495303872	3.739764017898	3.030964070213
7	2,414,682,040,998	2.954683814183	3.804357069772	3.164564560463

Table 1. Maxima over all monotone functions on n variables.

For $n = 6$ the maximum of $\min_i E_i/\mathrm{Inf}_i$ is attained by MAJ_3 of three AND_2 on disjoint pairs and by its dual, with every influence 3/16; the maximum of H/I for $n \leq 7$ by AND_n and OR_n. For $n = 7$ the maximiser of $\min_i E_i/\mathrm{Inf}_i$ has influences 0.09375 at three variables and 0.15625 at four, with ratios 3.0316 and 2.9547; its smallest ratio is at a coordinate of largest influence. Iterated compositions of monotone gates reach 4.506 [9], so the values in the table are far from the supremum over all monotone functions.

Table 2 gives, for $n \leq 6$, the largest E_i/Inf_i at the coordinate chosen by a rule, with ties broken against the rule. The coordinate of largest influence never does worse than $G_{\mathrm{mon}}(n)$ here, and at $n = 7$ it does not either, with ties broken in its favour. The coordinate of smallest positive influence reaches 6.2425 at $n = 7$ even with ties broken in its favour.

n	largest Inf_i	smallest Inf_i	largest $W_i^{(2)}/\mathrm{Inf}_i$	smallest $W_i^{(2)}/\mathrm{Inf}_i$
3	2.672488983973	2.672488983973	2.672488983973	2.672488983973
4	2.672488983973	3.523111966872	3.523111966872	3.523111966872
5	2.695887520153	4.335249356685	4.335249356685	4.335249356685
6	2.944495303872	5.019930166880	4.833940887931	5.019930166880

Table 2. Worst ratio at the coordinate selected by each rule, over all monotone functions on n variables.

The program `rule_minmax.py` computes $A(f)$, $B(f)$ and every ratio for all monotone functions on $n \leq 6$ variables. The largest value over f of $\min_{i \in A \cap B} E_i/\mathrm{Inf}_i$ is 2, 2.672488983973,

2.672488983973, 2.695887520153 and 2.944495303872 for $n = 2, \dots, 6$, which is $G_{\mathrm{mon}}(n)$ in each case. The rule does not always find the best coordinate: the minimum over all coordinates is not attained in $A \cap B$ for 8 of the 168 functions with $n = 4$, 190 of the 7,581 with $n = 5$ and 335,144 of the 7,828,354 with $n = 6$, and the quotient of the two minima reaches 1.108, 1.458 and 2.363. Taking the worst variable of $A \cap B$ instead gives 3.344977967946 at $n = 5$ and 3.650196658624 at $n = 6$; the worst variable of A, of B or of $A \cup B$ gives 4.950362212005 at $n = 6$.

8. Numerical checks

The script `check_hidden.py` in the code repository computes exact Walsh–Hadamard transforms of f_k for odd $k \le 17$, of six substituted functions with blocks of sizes 1, 3 and 5, and of f'_k for even $k \le 18$, and checks Lemma 4, Proposition 6 against a direct evaluation of the erasure entropy at every variable, Corollary 7, the bounds of Theorem 1, each rule of Theorem 2, and Proposition 9 including monotonicity; it also evaluates the law of Proposition 13 exactly. The census programs are `census_mono.py` ($n \le 6$) and `mono_census.c` ($n = 7$). The script `check_xand.py` compares the formula of Proposition 10 with exact transforms of $x \wedge g$ for 300 random monotone g on at most nine variables (largest difference $1.3 \cdot 10^{-15}$) and evaluates it and the bound on a grid of p; `check_gk.py` computes A, B and every ratio of f_k and $\tilde{f}_k$ for odd $k \le 13$ and checks monotonicity. The script `forced_rule.py` builds $F_{k,t}$ for $(k,t) = (3,1), (5,1), (7,1), (9,1), (3,2), (5,2), (3,3)$, checks that $A \cap B = \{x\}$ and that the ratio at x equals the one computed from the biased expansion of $\tilde{f}_k$ (to 10^{-9}), and tabulates the latter for $k \le 17$.

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This work was done by multiple AI agents, Claude (Anthropic) and Codex (OpenAI), connected to Math Brain [13] and working together inside the Brain framework [15], a shared, persistent memory in which the agents record claims, proofs, failed attempts and methods, retrieve one another's results, coordinate who works on what, and audit each other's work. For more on the details behind the Brain and Math Brain, see the Math Brain paper [14], the Math Brain repository [13] and the Brain framework repository [15].

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Declaration of competing interest

The author holds equity in Heights Labs and in Convex Nexus Capital and has a management role at Convex Nexus Capital. These entities are included in the funding declaration above.

Declaration of generative AI and AI-assisted technologies

The paper comes out of Math Brain, an ongoing project on the use of AI systems for research mathematics, and its results were developed with the Brain [15] and Math Brain [13]. In November 2026 a Claude Fable agent (Anthropic) and Codex agents running OpenAI's GPT-6.1, connected to the Brain and working on the same problem, studied the conjecture for monotone functions; a Codex agent found the constructions of Sections 3–6 other than Propositions 10, 11 and 12, with their proofs, and computed the census for $n \le 6$, and a Claude agent computed

the census for $n = 7$. Claude Opus 5.5 (Anthropic) found Propositions 10, 11 and 12 and the census of the rule of Section 5 built on shortest minterms and maxterms, rederived each proof, wrote the numerical checks and a second census program for $n \leq 6$, prepared the manuscript and audited it. These checks are not peer review. The author directed the work and is responsible for the content of the paper and for decisions concerning its dissemination.

Code and data availability

Code and outputs: `https://github.com/yspennstate/fei-hidden-coordinates`.

The paper is part of the Math Brain project, whose public repository is `https://github.com/yspennstate/math-brain-public` [13].

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